La Consolacion University Philippines
Valenzuela St., Capitol View Park Subdivision, Bulihan, City of Malolos, Bulacan

SCHOOL YEAR 2025-2026

COLLEGE OF INFORMATION TECHNOLOGY AND ENGINEERING

I. COURSE CODE : IT 202

II. COURSE TITLE : Computer Programming 1

III. COURSE DESCRIPTION :

This course introduces students to the vital role of computers and computer programmers in shaping modern society. It aims to foster a strong foundation in logical formulation techniques essential for programming, while promoting the development of good coding practices. Students will be taught the fundamentals of Java programming and explore significant real-world applications.

In line with La Consolacion University Philippines' commitment to forming **Christo-centric Catholic professionals**, the course integrates **Marian Augustinian values** in each lesson, ensuring that students not only achieve intellectual growth but also cultivate moral and ethical responsibility in the digital age. This holistic approach nurtures individuals who contribute positively to society, embodying **community, humility, and love of service**, as exemplified by the Blessed Virgin Mary and Saint Augustine. By the end of the course, students will have created their own Java applications, demonstrating both technical proficiency and a sense of purpose rooted in their Catholic faith.

IV. PRE-REQUISITE : IT 101

V. CREDIT UNITS : 2 units lecture / 1 unit laboratory

VI. COURSE LEARNING OUTCOMES:

Vision	Mission

Moved by the Spirit of the Risen Christ, La Consolacion University Philippines, envisions itself to become a catalyst for the wholistic transformational development of Catholic Augustinian Marian graduates rooted in Gospel values.	To achieve the above vision, LCUP commits itself to generate and make available to its Augustinian Marian students, programs and services that will make them: ● *exhibit academic and technical competence in their respective areas of discipline in the local and international environment;* ● *demonstrate their knowledge, critical thinking and creativity by exploring the various avenues of learning and searching for truth;* ● *manifest their willingness to serve specially the marginalized sector of our society;* ● *demonstrate productivity in various areas of learning as responsible citizens of the society; and* ● *live the Gospel values in the varied aspects of their personal and professional lives*

EXPECTED GRADUATE ATTRIBUTES	PROGRAM OUTCOMES	COURSE LEARNING OUTCOMES
EGA 1: Critical and Creative thinker EGA 2: Competent Catholic Augustinian-Marian Professional EGA 3: Socially Responsive Steward EGA 4: Transformative Lifelong learner	PO1. Articulate and discuss the latest developments in the specific field of practice; PO2. Effectively communicate orally and in writing using both English and Filipino; PO3. Work effectively and independently in multi-disciplinary and multi-cultural teams; PO4. Act in recognition of professional, social ethical responsibility; PO5. Preserve and promote "Filipino historical and cultural heritage; PO6. Analyze complex problems, and identify and define the computing requirements needed to design and appropriate solution;	CLO1. Define and analyze the computing requirements needed for an effective solution that is essential for developing holistic and impactful solutions. [EGA 1,3,4] [PO 1,2,3,6,8] CLO2: □ Incorporate ethical standards and social responsibility into technical designs, ensuring that solutions serve the common good, in line with Marian and Augustinian principles of truth and compassion. □ [EGA 1,3,4] [PO 1,2,3,6,8] CLO3: Collaborate effectively within multi-disciplinary and culturally diverse teams to develop Java-

	PO7. Apply computing and other knowledge domains to address real-world problems; PO8. Design and develop computing solutions using a system-level perspective; PO9. Utilize modern computing tools.	based programming solutions that address social challenges, demonstrating social responsibility and cultural sensitivity. [EGA 1,2,3,4] [PO 1,2,3,4,6,8,9] CLO4. Demonstrate the Filipino-Catholic Augustinian Spirituality through value webbing. [EGA 2] [PO 4,5]

VII. LEARNING PLAN:

SCHEDULE	INTENDED LEARNING OUTCOMES	COURSE CONTENTS	SUGGESTED LEARNING ACTIVITIES	SUGGESTED ASSESSMENT TASKS	EXPECTED VALUE/S TO BE INTEGRATED
Week1	ILO 1: Demonstrate a clear understanding of the course structure, expectations, and key programming concepts in Java, including how these skills align with real-world IT challenges and professional ethical standards. [CLO1]	**I. Orientation** A. Introduction to the Course Contents B. Course Requirements and Activities C. Course Objectives Grading System and Policies on Absences, Tardiness and Examinations	• Setting and sharing of Expectations from the Course • Discussion on the Vision, Mission and Goals of LCUP and of CITE • Presentation of Course Objectives and Contents, Course Requirements and Activities Review of Policies on Absences, Tardiness,	• Statement of Expectations • Review of Course Contents, Objectives, Activities and Assessment Tools • Review of Policies on Absences, Tardiness, Examinations, Grading System	• Compassion • Courage • Open mindedness • Respect • Obedience

			Examinations, Grading System		
Week 2	LO1. Explain the basic concepts and principles of Java programming, [CLO 1]	**II. Introduction to Java Programming** • A Brief History of Java language • Characteristics of Java Language • Reasons why programmers use Java language Features of Java Programming	• Lecture-Discussion/Recitation using Power Point Presentation • Video Presentation • Discovery Learning • Functioning Knowledge	• Brain Storming • Quiz • Assignment	• Self Awareness • Active participation and involvement • Courage • Humility • Analytical Skills
Week 3-4	ILO1: Demonstrate the skills by knowing the syntax and semantics of Java. of Language. [CLO 1,2] ILO2: Ensures that students gain a foundational understanding of Java and are able to apply basic programming concepts through hands-on practice. (CLO 2] [CLO 3]	**III. The Basic Input/Output Statements of Java Console** • The Basic Input/Output • Variable Declaration • Rules in declaring variables, ➢ Constants ➢ Operators • Java Installation Java Eclipse GUI	• Discussion using PowerPoint Presentation • Discovery Learning • Functional Knowledge • Research Study • Demonstration	• Recitation • Group Participation • Quiz • Machine Problem • Assignment	• Interiority • Work Ethics • Interpersonal Skills • Courage • Critical Thinking • Courage •

Week 4-5	ILO 1: Inference read and write data using Java streams [CLO 1] [CLO 2] ILO 2 Acquire the necessary skills and understanding of java methods. ILO4: Demonstrate the skills by knowing the syntax and semantics of Java. of Language. [CLO 3]	**IV. Java Libraries Program** • Parts of a Java program • The print and println function Data type • Int (Integer) • Float • Double • Char • Reasons to declare variables	• Lecture-Discussion using Power Point presentation • Seatwork • Laboratory • Machine Problem Video Presentation	• Recitation • Quiz Simulation of the external and internal parts of the System Unit	• Understanding • Exactness • Patience • Critical Thinking • Active participation and involvement • Critical Thinking • Patience • Appreciation
Week 6	Review topics covered by the examination	**PRELIM EXAMINATION**	Examination on the topics discussed during the Prelim Term	Result of the Examination	• Skills and Competencies • Critical Thinking
Week 7-8	ILO3: Understand the fundamentals of structured logic using decisions, loops, and array manipulation. [CLO 1] [CLO3] ILO4: Be able to apply conditional statements in Java to control program flow based on specific conditions, and write programs	**V. . Conditional Statements** • The Conditional Statements Syntax • If Statement • If Else Statement Switch/case Language	• Discussion using PowerPoint Presentation • Video Presentation • Laboratory Exercises: Machine Language Group Dynamics	• Recitation • Seatwork on the topics discussed • Laboratory Exercises	• Patience • Humility • Cooperation • Accuracy • Appreciation • Leadership Skills • Courage

	that make decisions dynamically depending on user input or computed values. [CLO 2] [CLO3]				
Week 9-11	ILO1: Utilize various looping constructs (for, while, do-while) in Java to execute repeated tasks, and design programs that efficiently handle repetitive processes and iterations, incorporating loops to solve real-world problems. CLO1 CLO2, CLO3 ILO2. Inference the structured programming concepts aCLOng with the Java syntax to implement them.	**XI. Looping Statements** • Java Loops ▪ For Loops ▪ Do While Loop ▪ Break/Continue	• Discussion using PowerPoint Presentation • Functioning Knowledge • Laboratory Exercises.	• Recitation • Group Participation • Quiz • Machine Problem	• Critical Thinking • Understanding • Precision • Accuracy • Appreciation • Humility
Week 12	Review topics covered by the examination	**MIDTERM EXAMINATION**	Examination on the topics discussed during the Midterm Term	Result of the Examination	• Skills and Competencies • Critical Thinking
Week 13	ILO4: declare, initialize, and manipulate arrays in Java, effectively using one-	**X. . Array** • Array • Example of Array • Two-Dimensional Arrays syntax:	• Discussion using Power Point Presentation	• Recitation • Quiz	• Active participation and involvement

	dimensional and multi-dimensional arrays to store, retrieve, and process data in various applications [CLO1] [CLO2] ILO6: Synthesize skills in De-bugging Java Language program.		• Functioning knowledge • Discovery Learning • Quiz on the discussed topic Laboratory Exercises	• Machine problem: Hands-on exercises using Java Console programs	• Understanding • Compassion Critical Thinking
Week 14	ILO1: Apply various string manipulation methods in Java. [CLO 1-2] . LO2: Synthesize skills in De-bugging Java Language program. [CLO2-3]	**XIII. Strings** • String Manipulation ➢ Length() ➢ charAt() ➢ indexOf() ➢ concat() ➢ replace ➢ ignorecase() ➢ substring()	Lecture: • Discussion using Powerpoint Presentation • Quiz • Laboratory: :Exercises: ➢ Machine Problem • Seatwork using trial and error discovery learning	• Recitation • Quiz • Seatwork-Trial and Error for discovery learning • Accurate Program output	• Critical Thinking • Active participation and involvement • Open mindedness • Humility • Patience • Courage
Week 15	ILO4: Demonstrate the skills by knowing the syntax and semantics of Java. [CLO1-2] ILO5: Build robust applications using Java console code. [CLO 1-2] ILO6: Synthesize skills in De-bugging	**XII. JAVA METHOD** • Java Parameters • Object Connection Java Recursion	• Discussion using Power Point Presentation • Functioning knowledge • Blended Learning • Quiz on the discussed topic • Laboratory Exercises	• Recitation • Quiz • Machine problem: Hands-on exercises using C programs	• Active participation and involvement • Understanding • Interiority • Courage • Critical Thinking

	Java Language program. [CLO2-3]				
Week 16-17	ILO1 Retreve data from a relational database using java class. [CLO 2-3] Demonstrate the Filipino-Catholic Augustinian Spirituality through value webbing.	X. **JAVA DATABASE CONNECTION** • Improvised Database UCanAccess Connection	• Discussion using Power Point Presentation • Quiz on the discussed topic • Cooperative Learning • Group Dynamics Laboratory Exercises	• Recitation • Quiz • Groupings • Machine problem: Hands-on exercises using Java programs	• Active participation and involvement • Understanding • Teamwork Skills • Courage Critical Thinking
Week 18	Review topics covered by the examination	**FINALS EXAMINATION**	Examination on the topics discussed during the Final Term	Result of the Examination	• Honesty • Integrity Critical Thinking

VIII. COURSE REQUIREMENTS :

1. Class attendance
2. Recitations
3. Quizzes
4. Examinations
5. Machine Problem : Laboratory Exercises

IX. REFERENCES :

A. BOOKS

Burd, B. (2022). *Java for dummies (Eighth edition).* John Wiley & Sons, Inc. QA 76.73 J83 B87 2022

Bernard, M. (2022). *Future skills: the 20 skills and competencies everyone needs to succeed in a digital world.* John Wiley & Sons, Inc. T 65.3 M37 2022

Farrell, J. (2019). *Java programming.* Cengage. QA 76.73 F37 2019

B. INTERNET REFERENCES

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Learn Java | Codecademy. https://www.codecademy.com/learn/learn-java. Accessed 19 June. 2024.

Learn Java Programming. https://www.programiz.com/java-programming. Accessed 16 February 2023

Java Tutorial - Tutorialspoint. https://www.tutorialspoint.com/java/index.htm. Accessed 29 December. 2023

Java tutorial: Learn Java Programming with examples. https://beginnersbook.com/java-tutorial-for-beginners-with-example. Accessed 15, March, 2023.

C. JOURNALS

Kok, A. J. et. Al., (2024). "*Analysis and Evaluation of a Searchable Exercise Repository for Training Java Programming",* Informatics in Education, 2024. DOI: 10.15388/infedu.2024.17

Park J. et. Al., (2023). "*Static Analysis of JNI Programs via Binary Decompilation*"**:**IEEE Transactions on Software Engineering. https://doi.org/10.1109/tse.2023.3241639

Syaifudin, Y.W., Funabiki, N., Kuribayashi, M. and Kao, W.-C. (2019), "A proposal of Android Programming Learning Assistant System with implementation of basic application learning", *International Journal of Web Information Systems*, Vol. 16 No. 1, pp. 115-135. https://doi.org/10.1108/IJWIS-08-2019-0038

D. ELECTRONICS

Java Eclipse IDE – Console/NetBeans

Quizizz

X.	**GRADING SYSTEM**	**:**
1.	Class Standing Quiz/Recitation Machine Problem Assignment	60%
2.	Examination	40 %
3.	Term Grade	CS (60%) + TE (40%)
4.	Final Rating	Prelim Grade + Midterm Grade + Final Grade

3

Prepared by:	Checked and Reviewed by:	Approved by:
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Revision Date:	
Modifications: __Learning Outcomes __Course Contents __Suggested Learning Activities __Suggested Assessment __Expected Values to be Integrated __Course Requirements __References __Grading System	